Defocus Deblurring at Native Sensor Resolution: Recovering Real Detail from In-Focus Evidence

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Abstract

Single-image defocus deblurring is developed and evaluated at about 2 MP, while cameras capture 24–200 MP. We study the problem at the native resolution of the sensor and find that the two obvious ways to bridge the gap both fail. Applying a state-of-the-art deblurring network directly to 30 MP images changes almost nothing, because the circles of confusion grow with resolution beyond anything seen in training. Deblurring a 4× downscaled image and upsampling the result recovers structure but loses native detail: with bicubic upsampling it is the most faithful pipeline we evaluate, yet perceptually worse than the blurry input. Generative super-resolution brings detail back by inventing it, costing up to 1.3 dB against bicubic upsampling of the same result and damaging regions that were already sharp. Across 5 deblurring networks, 7 upsamplers including three one-step diffusion models, training-free fusions and reference-based super-resolution, no baseline improves on bicubic upsampling of the same low-resolution result perceptually without losing fidelity. We observe that the missing detail is often present elsewhere in the same photograph: defocus is spatially varying, so the in-focus regions of the native observation contain real texture of the very materials that are blurred elsewhere. We propose METHOD*, a blur-guided upsampler that lifts any low-resolution deblurring method to native resolution. It keeps the structure of the low-resolution result exactly by construction and adds only native detail: taken from the observation where it is sharp or mildly defocused, and, where defocus has erased it, transferred from relevant in-focus exemplars retrieved anywhere in the image through a global memory, so that the patch-wise inference is not limited by its tile. To evaluate at native resolution, we rebuild the DPDD benchmark at 6720 × 4480 from the raw captures and define a protocol with standard metrics, registration and blur-stratified diagnostics. Results sentence:* METHOD *improves N methods by X while not altering in-focus regions.*

1. Introduction

Defocus blur is an unavoidable consequence of finite depth of field. Wide apertures are used to gather light, to freeze motion, or for subject isolation, and whenever the scene spans more depth than the lens can keep sharp, regions away from the focal plane are imaged through circles of confusion (CoC) whose size grows with their distance from that plane. Single-image defocus deblurring aims to recover the all-in-focus image and has progressed rapidly with learned models [1, 27, 38, 40, 41, 47, 70].

These models are developed at a resolution far below that of the images they are meant for. Interchangeable-lens cameras record 24–60 MP and smartphone main sensors 50–200 MP, but the standard DPDD benchmark [1] is distributed and evaluated at 1680 × 1120, although it was captured at the 6720 × 4480 resolution of a Canon EOS 5D Mark IV. Downscaling by a factor s shrinks every CoC by s, so the benchmark poses a much easier problem than the photograph the user took: blur kernels are small relative to the network's receptive field and training crops, and the finest detail, which is the hardest to restore, has already been discarded. Methods trained on larger images [44] and ultra-high-definition restoration methods (Sec. 2) exist, and recent captures reach 6000 × 4000, but they are still released and evaluated at a fraction of that resolution [45]; to our knowledge no defocus deblurring method has been evaluated against ground truth at the native resolution of the sensor, nor is it known how existing ones behave there.

We find that the two obvious ways to apply existing methods at native resolution both fail, for different reasons (Fig. 1, Sec. 4). *Native-resolution inference*, necessarily in tiles, changes almost nothing: on the 76 DPDD test scenes at 30 MP, four recent deblurring networks stay within 0.2 dB of the blurry input and the fifth [44] gains 0.6 dB, while the same networks gain 1.4–2.4 dB when the image is downscaled by 4. Tile size and test-time statistics correction [10] make no difference. The cause is not missing context or tile seams but blur of an extent the networks never saw. *Deblurring at low resolution and upsampling* recovers the

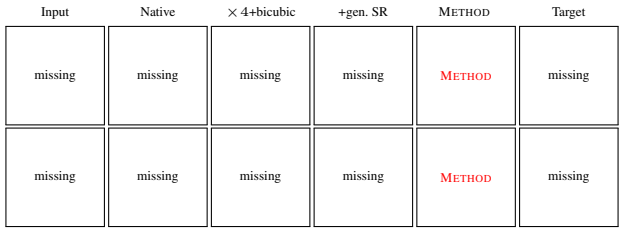


Figure 1. **Defocus deblurring at native resolution (30 MP DPDD).** Running a deblurring network directly at native resolution leaves the image as blurry as the input. Deblurring at 4× lower resolution and upsampling recovers structure but no native detail; generative super-resolution adds detail by invention and rewrites text and in-focus texture. METHOD copies the observation where it is sharp and transfers texture from in-focus exemplars of the same image where it is not. Crops: 256 px native windows of the lockers (1P0A2030) and the poster caption (1P0A1890); generative SR is OSEDiff on the ×4 DRBNet anchor. METHOD column.

structure (+1.5–1.7 dB over the input with bicubic upsampling, depending on the deblurring network) but none of the native detail, and the result is visibly soft. Restoring that detail with super-resolution is not a remedy: learned upsamplers add nothing over bicubic in fidelity, and generative ones [53, 62, 69, 71] invent detail, rewriting text into different glyphs and degrading regions that were already in focus (in the focal plane, −1.0 to −1.1 dB with a DRBNet anchor and up to −1.6 dB with a Restormer anchor, for one-step diffusion upsamplers built on text-to-image models). A conservative one-step model [64] avoids the damage but stays close to bicubic. All baselines we evaluate, including training-free fusions of the anchor with the observation and reference-based super-resolution, lie on one trade-off curve (Fig. 4): the most faithful pipeline is perceptually worse than the blurry input, and every perceptual gain is paid for in fidelity.

Our key observation is that the native-resolution observation is far from uninformative, and that what it lacks locally it often contains elsewhere. Defocus varies across the image: in-focus regions retain their full detail, mildly defocused regions retain part of it, and only strongly defocused regions have lost it. Photographs are also highly self-similar [16, 78]: the wall, fabric or foliage that is blurred in the background frequently appears sharp nearer the focal plane. A deblurred low-resolution image tells us *what* is where; the in-focus parts of the native observation tell us *what it looks like* at full resolution. Even a non-learned version of this idea, which pastes the high band of the best-matching in-focus patch into defocused regions, improves perceptual quality at almost no fidelity cost (Sec. 4.5).

We build on this observation with METHOD, a blur-guided upsampler that lifts any low-resolution deblurring method to native resolution. It takes the native blurry image, the low-resolution deblurred *anchor*, and a *blur map*, and works patch-wise. Three ideas shape the output. (i) An *anchor lock*: the network only adds content above the anchor's frequency band, so the result downscales exactly to the anchor and can never be less faithful than it at the benchmark resolution, unlike generative upsamplers. (ii) *The observation is used wherever it still holds detail*: guided by the blur map, the network reproduces it in focus and recovers the attenuated high frequencies under mild defocus, a regime that neither native deblurring nor low-resolution pipelines exploit. (iii) An *in-focus exemplar memory*, built once per image from sharp regions anywhere in the photograph, is queried by every patch through self-supervised features of the anchor, in which sharp and defocused regions are comparable; relevant exemplars supply native texture where defocus has erased it, and a learned "no match" option lets the model abstain. Because the memory is global, each patch draws on the whole image, and patch-wise inference is no longer limited by its tile. Because the upsampler depends on the low-resolution stage only through the anchor and the blur map, any existing deblurring method can be plugged in without retraining; the blur map can be taken from dual-pixel data when available or predicted from the anchor and the input otherwise.

Evaluating at native resolution raises its own difficulties. The full-resolution DPDD data exist only as raw captures, so we develop them and calibrate each capture to the official rendering (39 dB agreement at the benchmark resolution), register the f/22 targets to the f/4 inputs with a translation-only model (a homography over-fits the blur difference), and stratify all results by the local amount of defocus measured from the dual-pixel views. Headline metrics are standard and fixed before any result of our method was available; blur-stratified, in-focus-preservation and hallucination analyses explain the differences between methods.

In summary, our contributions are:

- An analysis of defocus deblurring at native sensor resolution, showing on 30 MP data that native inference is close to the identity and that low-resolution deblurring followed by (generative) super-resolution either loses or invents detail, and why.
- The observation that at native resolution the source of the missing detail depends on the local defocus (the observation itself in focus and under mild defocus, in-focus regions elsewhere in the same image under strong defocus), and that the two can be matched across focus in a deblurred low-resolution anchor. Its components (back-projection, feature modulation, patch retrieval) are known; we combine them into METHOD, an anchor-locked, blur-guided upsampler with a global in-focus exemplar memory that lifts any low-resolution method to native resolution. headline numbers: +X dB / DISTS /

human preference over the strongest baseline; N methods lifted, incl. held-out ones; no change in focal-plane regions.

- A native-resolution DPDD benchmark rebuilt from raw (76 test, 74 validation, 350 training scenes) with registration, dual-pixel blur stratification and a fixed protocol, and an evaluation of 5 deblurring networks, 7 upsamplers (three one-step diffusion models), 5 training-free fusions and reference-based super-resolution on it, which establishes the fidelity and perceptual ceilings of existing approaches.

2. Related Work

Defocus deblurring. Classical methods estimate a spatially varying defocus map, from edge cues [77] or learned predictors [26], and then deconvolve. Abuolaim and Brown [1] introduced DPDD, real pairs captured with a wide (f/4) and a narrow (f/22) aperture, together with DPDNet, which uses the dual-pixel (DP) views; later work modelled DP data further [2, 66]. Single-image methods design kernel-sharing atrous convolutions [47], predict inverse kernels in feature space [27], train on light-field-generated defocus [40], model blur with Gaussian kernel mixtures or recursive kernels [37, 38], use large kernels [41], or handle the misalignment of aperture-bracketed pairs [30]; general restoration backbones [8, 59, 70] are strong on DPDD as well. Bokehlicious [44] renders and removes bokeh with aperture control and is trained on multi-megapixel captures; the RealDefocus benchmark [45] is captured at 6000×4000 but trains at 2000×1500 and does not release or evaluate the captured resolution, and synthetic lens-diverse data [29] targets cross-camera generalisation. All of these are trained on crops and evaluated at about 2–3 MP, and hence calibrated to the CoC distribution at that resolution. We do not propose another low-resolution deblurrer; any of them serves as our anchor, and we address the step they leave open: the native-resolution result.

Ultra-high-definition restoration. UHD restoration methods for low-light enhancement, dehazing, deraining and motion deblurring [51, 54, 73, 76] typically process a strongly downscaled image and recover full resolution with learned upsampling, bilateral grids or frequency-domain operators. Our decomposition shares this coarse-to-fine spirit, but it is motivated by the growth of the CoC with resolution, uses an arbitrary existing method as the coarse stage, and, crucially, exploits the spatially varying amount of detail that survives in the defocused observation, both locally and in in-focus regions elsewhere, instead of predicting the fine scale from the coarse estimate alone.

Super-resolution and generative restoration. Regression-based super-resolution [9, 31] and GAN-based real-world models [56, 72] are natural upsamplers for a low-resolution deblurring result, and diffusion priors [39, 42, 60, 65] adapted from text-to-image models [32, 53, 63, 69] have raised perceptual quality further. Step distillation [43, 48, 68] has produced one-step restorers [57, 62, 71], and diffusion transformers [14, 36] provide stronger and more scalable priors, including one-step super-resolution without a text-to-image backbone [64]. These models are blind to the native observation: they generate detail consistent with the low-resolution input, which at native resolution is the anchor rather than the photograph. Our experiments show the consequence, invented text and texture and damaged in-focus regions, in line with the perception–distortion trade-off [5]. METHOD instead keeps the low-resolution result fixed and takes native detail from the observation itself; a generative prior, if used, is conditioned on the observation and restricted to regions where neither the observation nor the exemplars provide detail.

Reference-based and internal texture transfer. Reference-based super-resolution transfers texture from a separate high-resolution reference through patch correspondence [7, 21, 33, 67, 75], and retrieval-augmented restoration conditions large generative models on retrieved references [17, 25]. Closest to our setting, RefGC-SR^2 [50] upsamples a low-resolution generated image with a high-resolution reference, keeping the low frequencies of the generated image and transferring high-frequency statistics of the reference with a multi-step FLUX-Kontext model; it processes single 1024 px object images, and its reference is a separate, sharp photograph; reference-guided refinement of generated images [49] works at low resolution. Reference-based deblurring uses a sharp image of the same scene [79] or a reference generated from the blurry input itself [23], for motion blur. Internal learning exploits the recurrence of patches within a single image, across positions and scales [16, 46, 78], a principle shared with non-local filtering [6] and example-based texture synthesis [4, 13, 19]. Our exemplar memory combines both ideas in a setting where neither applies directly: the reference is the observation itself, its usable parts are defined by the local defocus, and correspondences are established in the deblurred anchor, where sharp and defocused regions look alike, while texture is taken from the native observation. Where the observation is in focus, the reference is pixel-aligned with the output and can be copied; the low band is locked exactly to the anchor; and a 30 MP image is processed in tiles that share one exemplar memory. We compare against reference-based super-resolution [7, 17, 25] given the observation as

reference.

Guided upsampling and patch-wise inference. Joint bilateral upsampling [24], the guided filter [18] and their learned counterparts [15, 61] transfer edges from a high-resolution guide to a low-resolution solution; they assume a guide that is sharp, whereas ours is sharp only where it is in focus. Patch-wise inference at high resolution is known to change the behaviour of restoration networks, partly through global statistics that TLC [10] corrects at test time, and tiled generative models fuse overlapping predictions or synchronize them to avoid seams [3, 12, 28]. At native resolution we find seams to be a minor effect compared with the blur-extent gap; the global exemplar memory gives every tile access to the whole image and makes the same material receive texture from the same sources across tiles.

3. Method

Given a native-resolution defocused image $x \in \mathbb{R}^{H\times W\times 3}$, an existing deblurring method $\mathcal{D}$ is applied to its $4\times$ downscaled version and yields the *anchor* $a = \mathcal{D}(x{\downarrow_4})$, which has the structure of the all-in-focus image but no native detail. METHOD produces the native all-in-focus estimate $\hat{y}$ from x, a and a blur map d giving the local amount of defocus (Fig. 2). It operates on 512×512 tiles T with overlap and blending, but every tile has access to a memory $\mathcal{M}$ of in-focus exemplars built once per image (Sec. 3.3).

3.1. Anchor-locked upsampling

The baselines show that the anchor's structure is worth keeping exactly: bicubic upsampling preserves it (the result, downscaled again, scores like the anchor itself), whereas generative upsamplers alter it and lose fidelity even at the anchor's resolution (Sec. 4.4). We therefore let the network contribute only what the anchor cannot express. With G the network, $a{\uparrow}$ the bicubically upsampled anchor, $\downarrow_4$ the area downscaling used throughout and $\uparrow$ an upsampling with $(e{\uparrow}){\downarrow_4} = e$ (bicubic, corrected by its own nearest-neighbour residual),

$$\begin{aligned} \Delta_T &= G\big(x_T,\, a{\uparrow}_T,\, d_T;\, \mathcal{M}\big), \\ \hat{y}_T &= a{\uparrow}_T + \Delta_T - (\Delta_T{\downarrow_4}){\uparrow}, \end{aligned} \tag{1}$$

so that $\hat{y}{\downarrow_4} = a$ exactly, as in one step of iterative back-projection [20]; after tiles are blended, the same projection is applied once more to the whole image. The output is as faithful as the anchor at the benchmark resolution by construction, any improvement of the anchor carries over directly, and the network is free to spend its capacity on the native high band. The price is that structural errors of the anchor are inherited; we leave correcting them to the low-resolution method.

3.2. Where native detail comes from

Defocus varies across the image, and with it the source of the missing high band. *In focus*, the observation already holds it: the network should reproduce x, lightly denoised (the f/22 targets are less noisy than the f/4 inputs). Under *mild and moderate defocus*, the circle of confusion attenuates but does not remove the frequencies above the anchor's band (the first zero of a disc blur of radius ρ native pixels lies at $0.61/\rho$ cycles per pixel, above the anchor's Nyquist frequency of $1/8$ for $\rho \lesssim 5$): the detail can be recovered from x itself by local deconvolution, a regime that neither native deblurring nor the low-resolution pipelines exploit. Under *strong defocus* it is gone, and must come from elsewhere: from in-focus exemplars of the same image when they exist (Sec. 3.3), and from the network's prior otherwise. G receives the blur map so that it can tell these regimes apart. We use the dual-pixel disparity directly as d, which is monotonic in the defocus and needs no calibration, only lightly smoothed so that depth edges stay sharp, together with its matching confidence, which separates "in focus" from "no texture to measure"; converting it to a circle-of-confusion radius or to per-band availability maps is evaluated as an ablation. When dual-pixel data are unavailable, d is predicted from the agreement between the anchor and the downscaled observation [77]. predicted-map head and its training.

3.3. In-focus exemplar memory

Inside a strongly defocused tile neither the observation nor the anchor carries native texture, but the same material is often in focus elsewhere in the photograph [16, 78]. *Features.* Defocused regions and their in-focus counterparts cannot be matched at native resolution, where one is blurred and the other sharp. In the anchor they are comparable, since the anchor is deblurred everywhere. We compute DINOv2 [35] features once on the whole anchor; at the anchor resolution each token covers about 56×56 native pixels. *Memory.* Keys are the anchor tokens of in-focus regions (small $|d|$), excluding flat ones; the corresponding values are the native observation crops under them, encoded at three scales to account for perspective. *Retrieval.* The tokens of the defocused part of a tile are the queries. For each, the K most similar keys by cosine similarity, their similarity scores and a learned null entry are passed to G, which attends to them,

$$r = \mathrm{softmax}\big(q^\top [k_{j_1}, \ldots, k_{j_K}, k_\varnothing]/\tau\big), \tag{2}$$

and receives the transferred features $\sum_i r_i v_{j_i}$. Retrieval therefore needs recall rather than precision; relevance is learned. The null entry lets G abstain and fall back to its prior where nothing relevant is in focus, and $\max_i r_i$ is a relevance map that we use for analysis. Because the memory is global, every tile draws on the whole image, and the result does not depend on the tile size. During training the memory is built from the whole image while the loss is computed

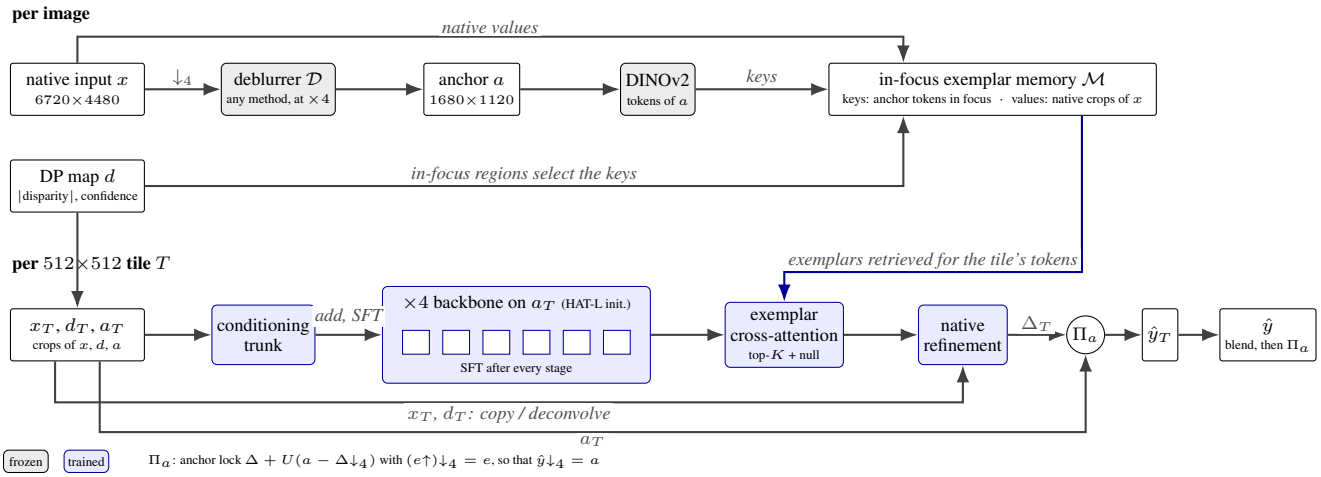


Figure 2. **Overview.** *Per image:* any deblurring method $\mathcal{D}$ produces the anchor a at $4\times$ lower resolution; DINOv2 features of the anchor index the in-focus regions (from the dual-pixel map d) into a memory whose values are native crops of the observation x. *Per tile:* the native crop x_T and d_T condition every stage of a $\times4$ network applied to the anchor tile; exemplars retrieved for the tile's tokens enter through cross-attention (with a null option); a native refinement sees x_T directly. The anchor lock Π_a keeps the result's $\times4$ band equal to the anchor, per tile and again after blending.

on a crop, and exemplars overlapping the crop are dropped so that the model learns to transfer rather than copy in place.

3.4. Network and training

Conditioning. The native input, rearranged onto the anchor grid, and d are encoded by a small trunk whose features modulate every stage of the network through spatial feature transforms [55], since the regime has to be known at every depth.

Network. G is a four-level U-Net. The two finest levels use convolutional NAFNet-style blocks [8]; at $1/4$ resolution channel attention [70] mixes information across the tile at linear cost; at $1/8$ resolution (64×64 tokens per tile) spatial self-attention provides tile-wide context and cross-attention reads the exemplar memory. sizes; whether a larger model initialized from a pretrained super-resolution network [9] is reported (used to validate the formulation).

Losses. An ℓ_1 loss on the full output dominates in focus. The f/22 target is slightly misregistered and has its own noise, so pixel losses alone would favor the mean and treat transferred texture as noise; in defocused regions we add an alignment-tolerant texture loss [34] and, in a second stage, a light adversarial loss. weights, schedule.

Generative variant. Where the memory abstains under strong defocus, a generative prior can fill in. We adapt a one-step super-resolution model [64], the only generative upsampler in our study that leaves in-focus regions intact, with LoRA, conditioned on x_T, d_T and the exemplar tokens, inside the anchor lock of Eq. (1), with the noise shared across tiles. only if the feed-forward model leaves a visible gap in such regions; otherwise an ablation.

3.5. Training data and plug-in anchors

We train on DPDD at native resolution (350 scenes, Sec. 4.1). The upsampler sees $\mathcal{D}$ only through the anchor, so any deblurring method can be plugged in at test time. Most published methods were trained on DPDD at exactly the anchor resolution, which makes their training-set anchors unrealistically good; Bokehlicious [44] was not, and provides our main training anchors. They are complemented by anchors of other methods and by simulated anchors (the downscaled target degraded by residual blur that grows with d, ringing and noise). Methods held out of training test the plug-in claim.

4. Experiments

4.1. Native-resolution DPDD

DPDD [1] contains 500 static scenes captured with a Canon EOS 5D Mark IV at f/4 (defocused input) and f/22 (target), split into $350/74/76$ scenes for training, validation and test. The released images are 1680×1120; the 6720×4480 originals exist only as raw captures. We develop every raw capture (libraw, camera white balance, the crop of the official framing, verified to 0.07 px) and calibrate it to the official rendering of the same capture with a per-capture colour polynomial, the converter's unsharp mask and a smooth local-tone field. After $4\times$ downscaling, our images agree

with the official ones to 38.8 dB (inputs) and 39.0 dB (targets) on held-out pixels before registration, and the input–target PSNR of our $\times 4$ level (24.24 dB on the 76 test scenes) is close to that of the official images (23.93 dB). Targets are registered to the inputs with a translation; a homography over-fits the blur difference between the two apertures (median 6.8 vs. 1.5 px estimated shift, $12/37$ vs. $0/37$ indoor scenes above 10 px). Validity masks and a 64 px border are excluded from all metrics. The left/right dual-pixel views of the input give a disparity proportional to the local defocus, which we smooth and use to stratify results into four bins (Fig. 3): focal plane (b_0, $|d| < 0.4$ px at 1680), mild, strong and very strong defocus (edges 1.5, 3.5 px); it correlates with the measured loss of sharpness between input and target (median Spearman 0.76). *Ground-truth sharpness.* At f/22 the diffraction cutoff of the camera (5.36 μm pixels, $\lambda = 550$ nm) is 0.44 cycles/px, and diffraction alone reduces the MTF to 0.65 at the Nyquist frequency of a $\times 4$ anchor (0.125 cycles/px) and to 0.21 at 0.3 cycles/px. The f/4 input is not free of loss either (aberrations, optical low-pass filter, demosaicking), and what matters for evaluation is whether the target holds the detail that the camera records in focus, which is the detail a method can restore or transfer. A paired measurement on 336 focal-plane edges of 37 scenes gives a target-to-input MTF ratio of 1.05, 1.24, 0.99 and 0.64 at 0.1, 0.2, 0.3 and 0.4 cycles/px (relative Gaussian blur $\sigma \approx 0$ px, 95% CI 0–0.31): the target is as sharp as the in-focus input up to 0.3 cycles/px and softer only near Nyquist. Plain PSNR/SSIM are therefore meaningful at native resolution, but they penalize correct detail above about 0.3 cycles/px. We report them as the fidelity criterion, together with a sensitivity variant in which predictions are blurred with the f/22 diffraction PSF before scoring numbers; suppl., and use DISTS and LPIPS to measure restored detail. All results use the 76 test scenes. Because our $\times 4$ level is a re-rendering, results at the benchmark resolution are reported both on it and on the original 1680×1120 images, on which our evaluation reproduces published numbers to within 0.06 dB; the two renderings shift individual models by -0.58 to $+0.16$ dB and change their order (Tab. 1).

4.2. Protocol

Metrics. We report standard full-reference metrics computed out of the box on the whole native image: PSNR, SSIM [58], LPIPS [74] and DISTS [11], the no-reference MUSIQ [22] and CLIPIQA [52], PSNR/SSIM after $4\times$ downscaling (the standard DPDD resolution, which makes native results comparable with published numbers), and runtime per 30 MP image. Blur-stratified PSNR and DISTS, the change in the focal-plane bin relative to the input, PSNR after a ± 2 px shift search per 512 px tile, and text crops serve as diagnostics computed identically for every method; they explain differences but are not used to rank. Every comparison is paired per image with 95% bootstrap confidence intervals. The protocol was fixed before any result of METHOD was available.

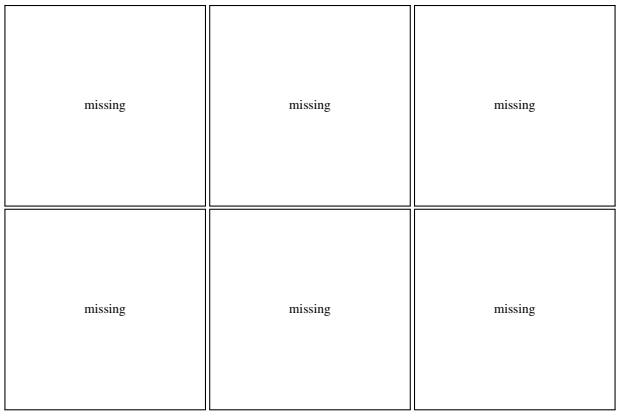


Figure 3. **Dual-pixel defocus bins** on three test scenes (`1P0A1002`, `1P0A2030`, `1P0A1890`); top: input at $\times 4$, bottom: bins overlaid (blue: focal plane b_0, green b_1, yellow b_2, red b_3). Flat, low-texture regions receive low dual-pixel confidence and can fall in b_0 even when defocused (the plain wall at the left of the first scene).

Inference. Each method runs as in its paper: methods evaluated on whole 1680×1120 images run whole at $\times 4$ and on 1120 px tiles at higher resolution; super-resolution models use their code's tile size in low-resolution pixels (SwinIR 400, HAT 512, OSEDiff and VOSR 128, *i.e.* their 512 px training size, S3Diff 192, its latent tile); overlap is $1/8$ of the tile for all, with linear blending, in one shared tiling implementation. Each tile follows the method's official one-step path; per-image quantities (S3Diff's degradation estimate) are computed on the whole image, and the sampling noise of the generative upsamplers is drawn once per image and shared by overlapping tiles.

4.3. Baselines

Native inference: Restormer [70] (with and without TLC [10]), LaKDNet-L [41], DRBNet [40], IFAN [27] and Bokehlicious [44], with released weights. *Low-resolution deblurring and upsampling*: an anchor deblurred at $\times 4$ (DRBNet or Bokehlicious, the strongest at that resolution; other anchors in the supplementary) followed by bicubic, real-world SwinIR [31], HAT-L and Real-HAT [9], or the one-step generative upsamplers OSEDiff [62], S3Diff [71] and VOSR [64]. *Training-free fusion* of the anchor and the native input using the dual-pixel blur map: a composite that copies the input in focus and the upsampled anchor elsewhere, a multi-scale composite that also uses a $\times 2$ deblur, a guided filter [18], adding the input's high band, and

Method	deblur at ×4				deblur at native	
	original		ours		ours	
	PSNR	Δ	PSNR	Δ	PSNR	Δ
Blurry input	23.93	–	24.24	–	23.06	–
Restormer [70]	26.04	+2.11	25.60	+1.37	22.99	−0.08
+ TLC [10]	–	–	–	–	22.99	−0.07
LaKDNet-L [41]	**26.31**	+2.38	25.74	+1.50	23.00	−0.06
DRBNet [40]	25.79	+1.86	25.85	+1.62	23.12	+0.06
IFAN [27]	25.42	+1.49	25.58	+1.35	23.22	+0.15
Bokehlicious [44]	26.13	+2.20	**26.08**	+1.84	**23.70**	+0.64

Table 1. **The resolution gap.** The same networks gain 1.4–2.4 dB over the input at the benchmark resolution and almost nothing at native resolution. PSNR in dB, each scored at the resolution it runs at; Δ: paired gain over the input. "original": the released 1680 × 1120 images (published numbers reproduced to 0.06 dB); "ours": our raw-built rendering, the ×4 level of the native set. 76 test scenes. Native 512 px instead of 1120 px tiles: ≤ 0.06 dB difference. which ×4 rendering leads is open; refresh from handoff 4.

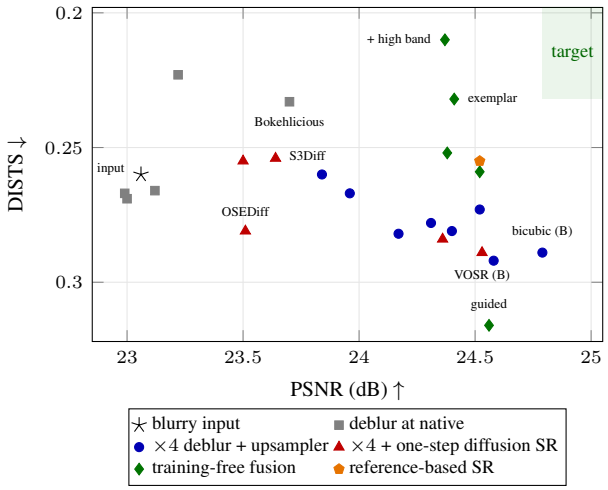


Figure 4. **Fidelity vs. perceptual quality of existing approaches** at native resolution (76 scenes; up and right is better). Every baseline lies on or below one trade-off curve between the most faithful pipeline (×4 Bokehlicious + bicubic, perceptually worse than the input) and fusions that add the input's high band. The shaded region, the fidelity of the best low-resolution pipeline with the texture of the non-learned exemplar transfer, is empty. (B): Bokehlicious anchor, otherwise DRBNet. METHOD points.

a non-learned version of our exemplar transfer; their parameters are tuned on the validation set. *Reference-based super-resolution*: DATSR [7] with a reference built per tile from the observation (the co-located crop and the three most similar in-focus crops). *Retrained at native resolution*: Restormer and DRBNet fine-tuned on native training crops. retrained rows.

4.4. Results

The resolution gap (Tab. 1). Every network that improves the input by 1.4–2.4 dB at the benchmark resolution is within 0.2 dB of the input at native resolution, except Bokehlicious, which is trained on multi-megapixel data and gains 0.6 dB. Changing the tile size from 1120 to 512 px changes the result by at most 0.06 dB, and TLC, which corrects global statistics for large inputs, changes nothing: the networks fail because the native blur is larger than anything they were trained on, not because of tiling.

Ceilings of existing approaches (Fig. 4). Four observations hold for both anchors and both ×4 renderings. (i) Native inference is bounded by the input: four of five networks change PSNR by less than 0.2 dB, and the best (Bokehlicious, +0.64 dB) remains 1.1 dB below the low-resolution pipelines. (ii) The fidelity ceiling is set by deblurring at ×4 and upsampling bicubically (24.79 dB with Bokehlicious); no learned or generative upsampler improves on bicubic in PSNR or SSIM, with either anchor. (iii) Perceptual gains are paid for in fidelity: the best LPIPS among upsamplers (S3Diff) costs 0.9 dB, the best DISTS overall (adding the input's high band) costs SSIM, and the text-to-image-based generative upsamplers damage the focal plane by 1.0–1.1 dB with the DRBNet anchor (Tab. 3) and by up to 1.6 dB with the Restormer anchor. (iv) Real texture helps: the non-learned exemplar transfer has a better DISTS than any generative upsampler (0.232 vs. ≥ 0.254), within 0.2 dB of bicubic upsampling of the same anchor and without damaging the focal plane. The region combining both, which no baseline reaches, is the target of METHOD.

Main comparison (Tab. 2). Deblurring at ×4 and upsampling bicubically gains 1.5–1.7 dB over the input (Bokehlicious anchor: 24.79 dB, the best PSNR and SSIM in the table), but it is smooth: its LPIPS and DISTS are worse than those of the blurry input. The perceptual upsamplers trade fidelity for texture: Real-HAT costs 0.4–0.5 dB and S3Diff 0.9–1.3 dB; S3Diff has the best LPIPS and no-reference scores of all upsamplers, and the only DISTS (0.254) better than that of the input. VOSR, a one-step diffusion transformer, is the most conservative generative upsampler: within 0.3 dB of bicubic, with better LPIPS and no-reference scores. The same ordering holds for both anchors. METHOD rows and the comparison sentence.

4.5. Analysis

Where detail is lost and where it is invented (Tab. 3). The gains of the low-resolution pipelines grow with the amount of defocus, as expected. In the focal plane, where the input is already sharp, the text-to-image-based gener-

	Method	native resolution PSNR↑	SSIM↑	LPIPS↓	DISTS↓	MUSIQ↑	CLIPIQA↑	at ×4 PSNR↑	SSIM↑
	Blurry input	23.06	0.596	0.370	0.260	27.3	0.321	24.24	0.711
native	Restormer [70]	22.99	0.601	0.382	0.267	28.9	0.317	24.14	0.713
native	LaKDNet-L [41]	23.00	0.575	0.390	0.269	29.2	0.325	24.33	0.711
native	DRBNet [40]	23.12	0.586	0.376	0.266	29.1	0.329	24.35	0.714
native	IFAN [27]	23.22	0.601	**0.348**	**0.223**	31.4	0.330	24.47	0.727
native	Bokehlicious [44]	23.70	0.642	0.369	0.233	31.8	0.335	24.89	0.758
DRBNet ×4	bicubic	24.58	0.662	0.573	0.292	24.5	0.408	25.87	0.778
DRBNet ×4	Real-HAT [9]	24.17	0.649	0.447	0.282	37.3	0.374	25.43	0.762
DRBNet ×4	S3Diff [71]	23.64	0.602	0.425	0.254	45.5	0.489	25.43	0.760
DRBNet ×4	VOSR [64]	24.36	0.649	0.510	0.284	40.8	0.450	25.80	0.778
Bokehl. ×4	bicubic	**24.79**	**0.667**	0.546	0.289	24.7	0.424	**26.10**	**0.801**
Bokehl. ×4	Real-HAT [9]	24.31	0.648	0.452	0.278	40.0	0.377	25.65	0.777
Bokehl. ×4	S3Diff [71]	23.50	0.607	0.427	0.255	**48.4**	**0.524**	25.31	0.775
Bokehl. ×4	VOSR [64]	24.53	0.658	0.522	0.289	40.1	0.459	25.93	0.797
	METHOD, Bokehlicious anchor	–							
	METHOD, DRBNet anchor	–							

Table 2. **Native-resolution DPDD (76 test scenes, 6720 × 4480).** Anchors are deblurred at ×4 on our rendering. "at ×4": the native result downscaled to the standard DPDD resolution. Further baselines (HAT-L, SwinIR-real, OSEDiff, training-free fusion, DATSR) in Fig. 4 and the supplementary. runtime column (warm timing pass); CIs in the supplementary.

Method	ΔPSNR vs. input by defocus bin: focal	mild	strong	v. strong	DISTS_{b_3}↓
Blurry input	0	0	0	0	0.261
DRBNet native	−0.02	+0.10	+0.13	+0.10	0.264
×4 + bicubic	+0.44	+1.32	+2.08	+2.17	0.310
×4 + HAT-L	+0.40	+1.23	+2.01	+2.08	0.297
×4 + SwinIR-real	−0.21	+0.70	+1.46	+1.72	0.303
×4 + Real-HAT	−0.05	+0.92	+1.71	+1.87	0.310
×4 + OSEDiff	**−1.11**	+0.26	+1.13	+1.48	0.315
×4 + S3Diff	−0.99	+0.43	+1.26	+1.60	–
×4 + VOSR	+0.08	+1.05	+1.87	+2.06	–
DP composite	+0.14	+1.31	+2.08	+2.17	0.308
exemplar (non-learned)	+0.14	+1.18	+1.92	+1.98	0.266
METHOD	–				

Table 3. **Where methods gain and lose** (76 scenes, DRBNet anchor). Mean paired PSNR change over the input in each dual-pixel defocus bin, and DISTS on 512 px tiles of the most defocused bin.

ative upsamplers are *worse than the input*: OSEDiff by 1.1 dB, with 87% of the scenes affected, and S3Diff by 1.0 dB; VOSR leaves the focal plane unchanged (+0.1 dB) but also gains less elsewhere than bicubic upsampling. Visually, OSEDiff rewrites caption text into different glyphs, alters faces and adds banding in dark regions, while SwinIR-real and Real-HAT add halos and contrast (Fig. 6). The bicubic pipelines slightly *gain* in the focal plane, because the f/22 target is less noisy than the f/4 input; PSNR thus rewards smoothing there, which is why we report in-focus preservation separately. Conversely, DISTS in the most defocused bin ranks the blurry input above every smooth pipeline: perceptual metrics at native resolution reward realistic grain and texture, which the smooth pipelines lack and which METHOD aims to restore from real evidence.

Qualitative comparison (Fig. 6). At native resolution the DPDD-trained network leaves the lockers and the poster caption as blurry as the input. Deblurring at ×4 and upsampling bicubically or with HAT-L recovers edges and legible text, but no detail beyond the anchor; the DP composite stays close to this result. The generative upsamplers add detail that is not the image's own: OSEDiff rewrites the caption into different glyphs and changes the shape of the locks, S3Diff sharpens the caption with altered glyphs and adds grain to the dark scene, SwinIR-real adds contrast and halos, and VOSR2 stays closest to bicubic.

In-focus exemplars carry the missing texture (Fig. 5). Within one image the same material often appears both in focus and defocused (Fig. 5): the defocused copy loses its fine texture at native resolution, deblurring at ×4 recovers its geometry but not the fine texture, and generative upsamplers replace it by texture that is not the material's own (a regular mesh for woven fabric). The in-focus copy is the only real evidence of that texture in the image. The non-learned exemplar transfer matches each defocused 32 px cell to the most similar in-focus cell of the same image at

Variant	native PSNR↑	native DISTS↓	ΔPSNR focal	ΔPSNR b_3	×4 PSNR↑
×4 + bicubic	24.79	0.289	+0.44	+2.43	26.10
METHOD (full)	–				
components					
w/o anchor lock	–				
w/o native input	–				
w/o blur map	–				
w/o exemplar memory	–				
exemplars					
random in-focus exemplars	–				
pixel features, not DINOv2	–				
oracle exemplars	–				
blur map					
CoC radius, not disparity	–				
per-band maps	–				
predicted, not dual-pixel	–				
training and inference					
simulated anchors only	–				
real anchors only	–				
tile 256 / 1024	–				
+ generative prior	–				

Table 4. **Ablations** (76 test scenes, Bokehlicious anchor). ΔPSNR: paired change over the input in the in-focus bin (focal) and the most defocused dual-pixel bin (b_3). fill from the ablation runs.

Anchor	×4 + bicubic PSNR↑	×4 + bicubic DISTS↓	METHOD PSNR↑	METHOD DISTS↓
Bokehlicious [44]	24.79	0.289	–	
DRBNet [40]	24.58	0.292	–	
Restormer [70]	24.48	0.319	–	
LaKDNet-L [41]	24.65	0.280	–	
IFAN [27]	24.40	0.296	–	

Table 5. **Plug-in.** METHOD on the anchors of different deblurring methods (76 scenes, native resolution). mark the anchors held out of training.

×4 and adds its native high band. In a pilot on four scenes, matching halved the PSNR cost and doubled the perceptual gain compared with copying random in-focus cells, and an oracle that selects cells using the target shows how much usable texture exists in focus. full-set numbers for random/oracle; relevance-map figure.

4.6. Ablations

Components (Tab. 4). Each ablation removes one source of information or one constraint. Without the anchor lock the network is free to alter the anchor's structure, which the ×4 column exposes. Without the native input the model becomes an anchor-only super-resolver and loses both the copy and the deconvolution regime. Without the blur map it cannot tell the regimes apart. Without the memory, texture under strong defocus must come from the network's prior alone. results.

Exemplars. Random in-focus exemplars test whether retrieval matters, pixel features test the choice of DINOv2 for matching across focus, and exemplars chosen with the target's high band bound what retrieval could achieve. results; retrieval recall against the oracle.

Blur map. The dual-pixel disparity is used as is; converting it to a circle-of-confusion radius or to per-band availability maps tests whether making the physics explicit helps. Replacing it by the predicted map measures the cost of not having dual-pixel data. results.

Training anchors and tiles. Training on simulated or on real anchors only isolates the role of the anchor mix. Because every tile reads the same global memory, the tile size should matter little. results; whether the generative prior adds anything in regions where the memory abstains.

Plug-in (Tab. 5). The upsampler sees the deblurring method only through its anchor. Applying it unchanged to the anchors of all five methods, including methods held out of training, tests whether the gains transfer and whether a better anchor yields a better result. results.

Human study. 2AFC preference against the strongest baselines on native-resolution crops.

5. Conclusion

Defocus deblurring at the resolution at which photographs are taken is not solved by existing methods: run natively they barely change the image, and run at low resolution they need an upsampler that either loses or invents detail. On the native DPDD benchmark every existing approach we evaluated lies on a single fidelity–perception trade-off, with bicubic upsampling of a low-resolution result as the fidelity ceiling and in-focus damage as the price of generative detail. Because defocus varies across the image, the photograph itself holds much of the detail that is missing, in its in-focus regions. METHOD uses this evidence to lift any low-resolution deblurring method to native resolution. results summary; limitations: dual-pixel oracle vs. predicted blur map, scenes without in-focus counterparts, runtime.

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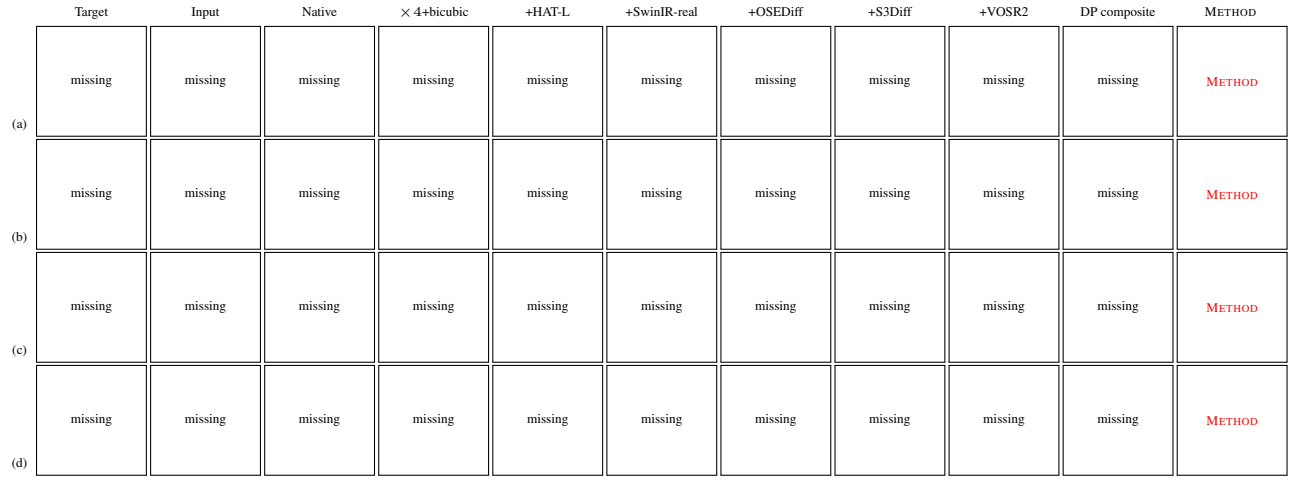


Figure 5. **The same texture in focus and defocused in one image, and how it evolves with deblurring and upsampling.** (a),(b) printed script on fabric (1P0A1046), (c),(d) woven fabric (1P0A1112); (a),(c) in focus, (b),(d) defocused (512 px native windows shown at half size, DRBNet anchor). Input and native inference keep the defocused texture blurred. ×4 deblur + bicubic or HAT-L recover the stroke geometry but not the fine weave. OSEDiff smooths the strokes and replaces the in-focus weave by a regular mesh. S3Diff adds weave-like texture to the defocused fabric; VOSR2 adds a regular grid at the right edge. The in-focus rows contain the texture that exemplar transfer copies into the defocused ones. METHOD column.

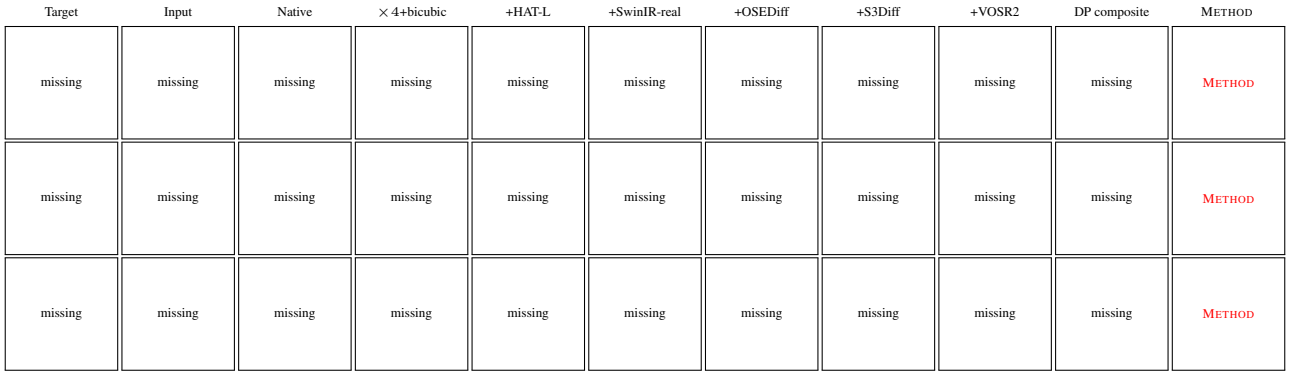


Figure 6. **Qualitative comparison at native resolution** (256 px native windows; ×4 rows use the DRBNet anchor). Native-resolution inference leaves the input as blurry as it is. ×4 deblur + bicubic and + HAT-L recover edges and legible text, and so does VOSR2, which stays close to bicubic. SwinIR-real adds contrast and alters glyph shapes; OSEDiff rewrites the poster caption into different glyphs, changes the shape of the locks and adds banding in the dark scene; S3Diff is sharper but also changes glyph shapes and adds grain in the dark scene. Rows: lockers (1P0A2030), poster caption (1P0A1890), dark scene (1P0A2202). METHOD column; remaining selected scenes.

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